

Drug-Drug Interactions Prediction

TEAM MEMBERS:

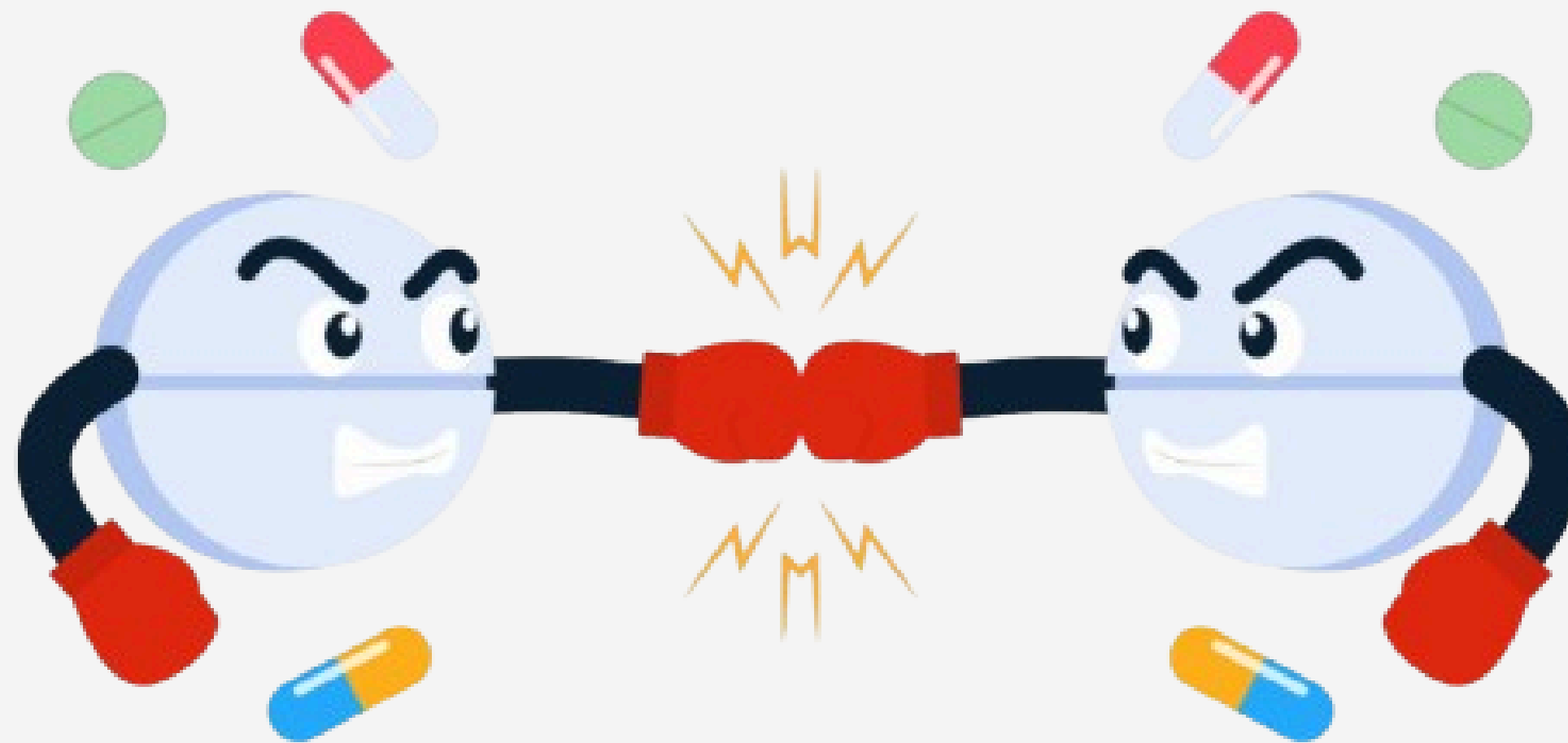
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What are DDIs?

A DDI occurs when one drug alters the effect of another, leading to toxicity or reduced efficacy.



The Problem

Polypharmacy

More people are taking 5+ medications simultaneously.

Search Space

Number of Approved Drugs: 5,000 - 7,000

Possible Pairs: 12.5 Million - 24.5 Million

! Impractical to experimentally examine
all possible drug pairs.

Dataset

DrugBank Multi-Typed DDI:

- 191,808 DDI pairs
- 1,706 drugs
- 86 interaction types

Drug1_ID	Drug1	Drug2_ID	Drug2	Y
DB04571	<chem>CC1=CC2=CC3=C(OC(=O)C=C3C)C(C)=C2O1</chem>	DB00460	<chem>COC(=O)CCC1=C2NC\C=C3/N=C/C=C4\N\C(=C/C5=N/C...</chem>	1
DB00855	<chem>NCC(=O)CCC(O)=O</chem>	DB00460	<chem>COC(=O)CCC1=C2NC\C=C3/N=C/C=C4\N\C(=C/C5=N/C...</chem>	1
DB09536	<chem>O=[Ti]=O</chem>	DB00460	<chem>COC(=O)CCC1=C2NC\C=C3/N=C/C=C4\N\C(=C/C5=N/C...</chem>	1
DB01600	<chem>CC(C(O)=O)C1=CC=C(S1)C(=O)C1=CC=CC=C1</chem>	DB00460	<chem>COC(=O)CCC1=C2NC\C=C3/N=C/C=C4\N\C(=C/C5=N/C...</chem>	1
DB09000	<chem>CC(CN(C)C)CN1C2=CC=CC=C2SC2=C1C=C(C=C2)C#N</chem>	DB00460	<chem>COC(=O)CCC1=C2NC\C=C3/N=C/C=C4\N\C(=C/C5=N/C...</chem>	1

Splitting Mechanism

Cold Splitting

- Partitions data based on unique identities (drug IDs).
- Divides the list of all unique drugs into mutually exclusive sets for training, validation, and testing.
- A drug-drug interaction pair is then only included in a specific split if both drugs in that pair belong to that split's assigned drug identities.

Cold splitting is done on Drug1_ID & Drug2_ID

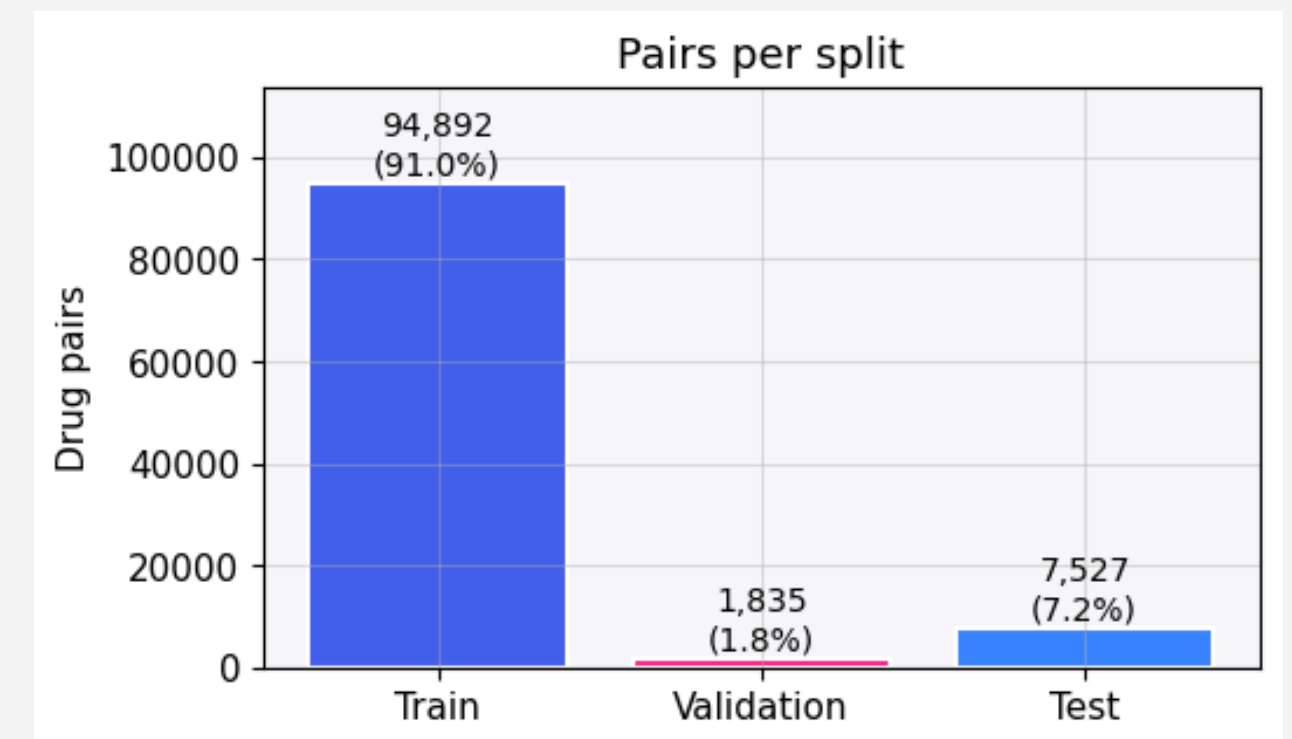
Train pairs: 94,892

Val pairs: 1,835

Test pairs: 7,527

Unique drugs: 1,646

Total pairs: 104,254



Baseline Models



Siamese GCN

processes each drug in total isolation

Input: raw molecular graphs

Encoding: Uses 3 layers of Graph Convolutional Networks (GCN)

Pooling: implemented to flatten the entire drug vectors (vector for each atom) into a single vector

Interaction: The two vectors are simply concatenated and fed to an MLP

Weaknesses:

- No Co-Attention: It doesn't look at how a specific part of Drug A reacts to a part of Drug B.
 - No Knowledge Graph: It is blind to biological targets like enzymes or pathways.
- 

Baseline Models



DeepDDI

represents drugs as fixed-length bit-vectors & uses predefined chemical rules

Input: Morgan Fingerprints (bit-vectors)

Architecture: A very deep Multi-Layer Perceptron with large hidden layers

Mechanism: It learns to map the combination of these two bit-vectors directly to one of the 86 interaction types.

Weaknesses:

- Not Graph-Aware: It doesn't understand the spatial arrangement of atoms
 - Struggle to generalize to entirely new chemical families
- 

Proposed Approach (KG-SS-GNN)

Knowledge-Graph-enhanced Substructure-aware Siamese GNN

1. Pre-processing

RDKit parses the SMILES string into a graph

BRICS Fragmentation

The fragments are converted to Morgan Fingerprints

2. GAT Encoder with BRICS Fusion

For each atom, it computes how much to attend to each of its neighbors.

Mechanism: It uses a 3-layer Graph Attention Network (GAT)

Siamese Setup: Both Drug A and Drug B pass through the same encoder

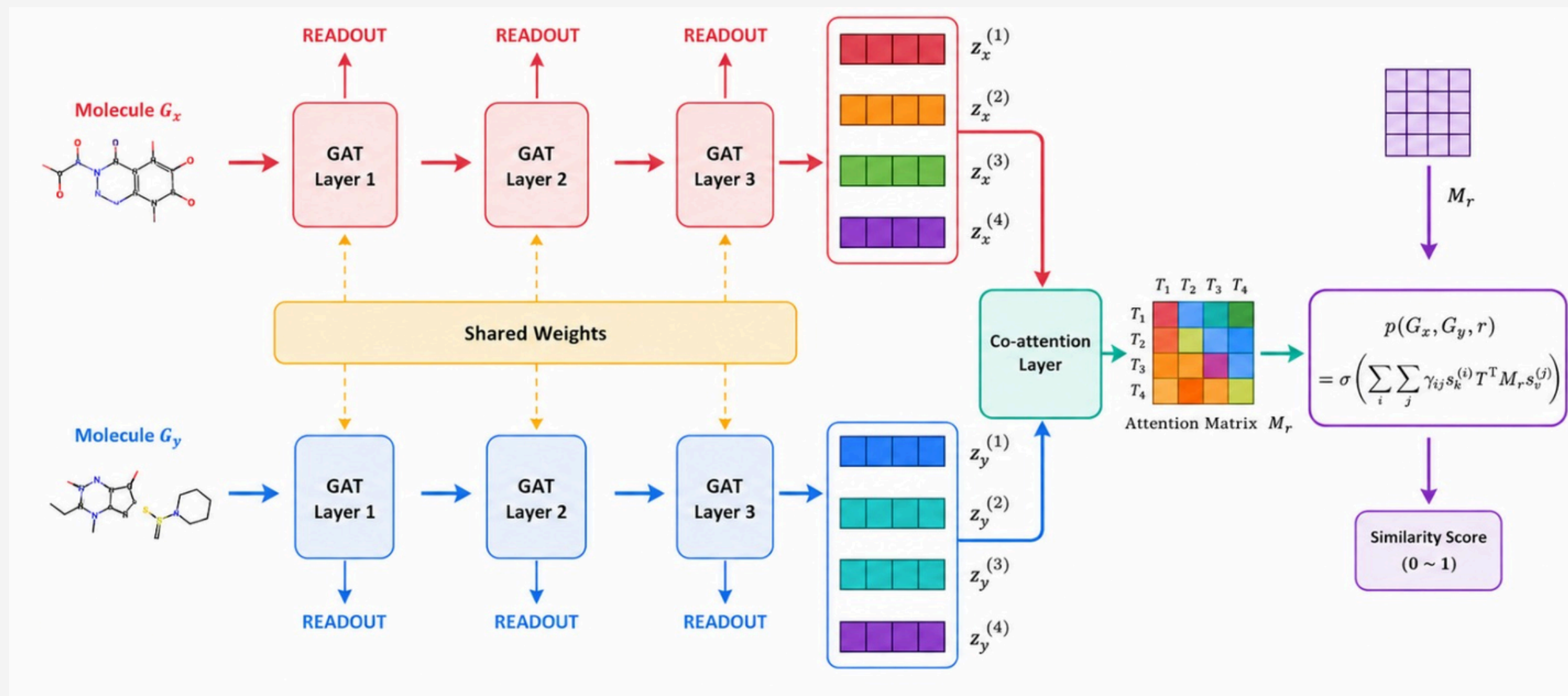
Fusing the global graph embedding with the substructure fingerprints before passing it to the Co-Attention layer

Proposed Approach (KG-SS-GNN)

Knowledge-Graph-enhanced Substructure-aware Siamese GNN

3. Cross-Drug Co-Attention

calculates a relevance score between the two drugs



Proposed Approach (KG-SS-GNN)

Knowledge-Graph-enhanced Substructure-aware Siamese GNN

4. Knowledge Graph (KG) Enhancement

builds a 141-dimensional biological profile for each drug:

64 bits: Radius-1 Morgan

64 bits: Radius-3 Morgan

13 bits: 13 Physical-Chemical Descriptors

5. The Joint Classifier (MLP)

the model concatenates five distinct representations:

1) Embedded features of Drug A & Embedded features of Drug B

2) The co-attended interaction features

3) KG features for Drug A & KG features for Drug B

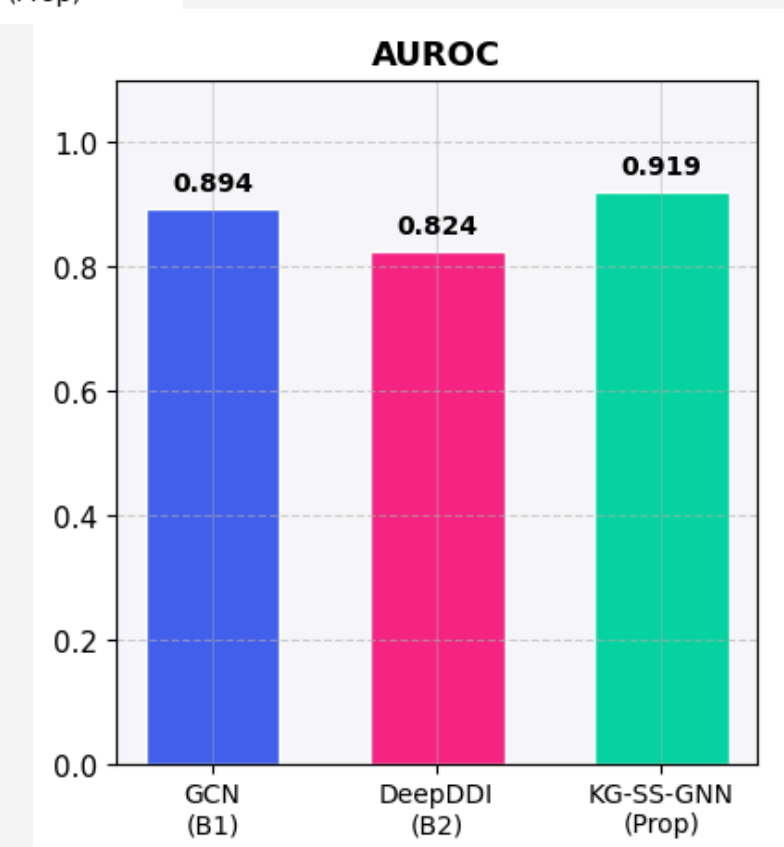
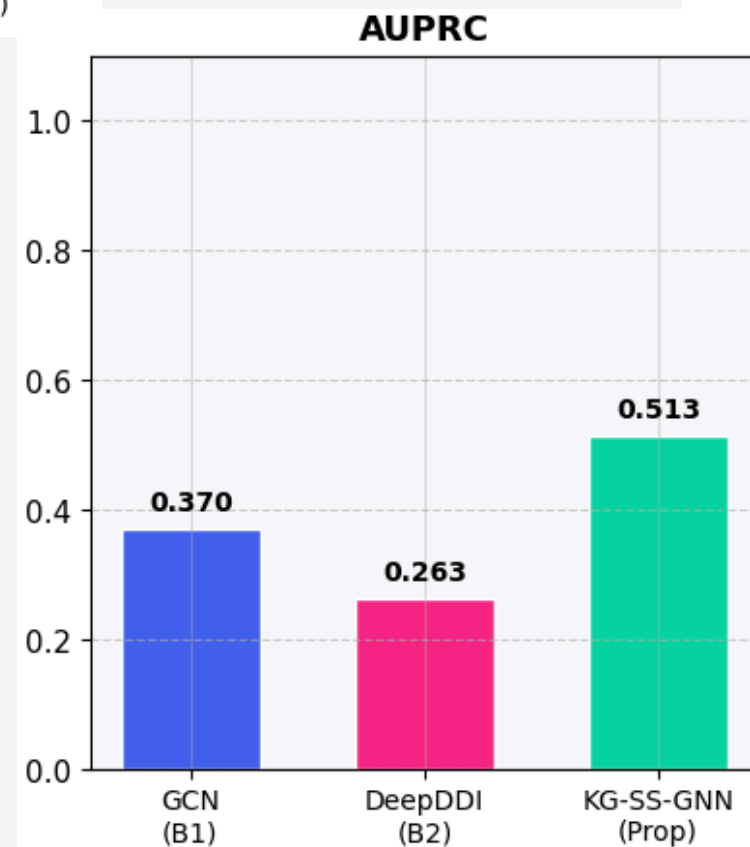
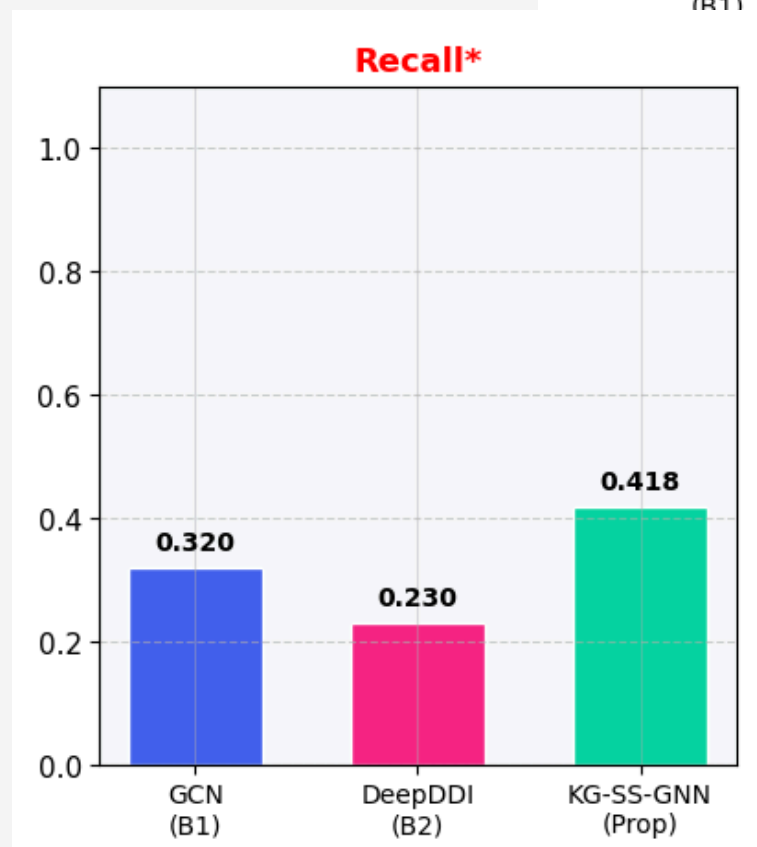
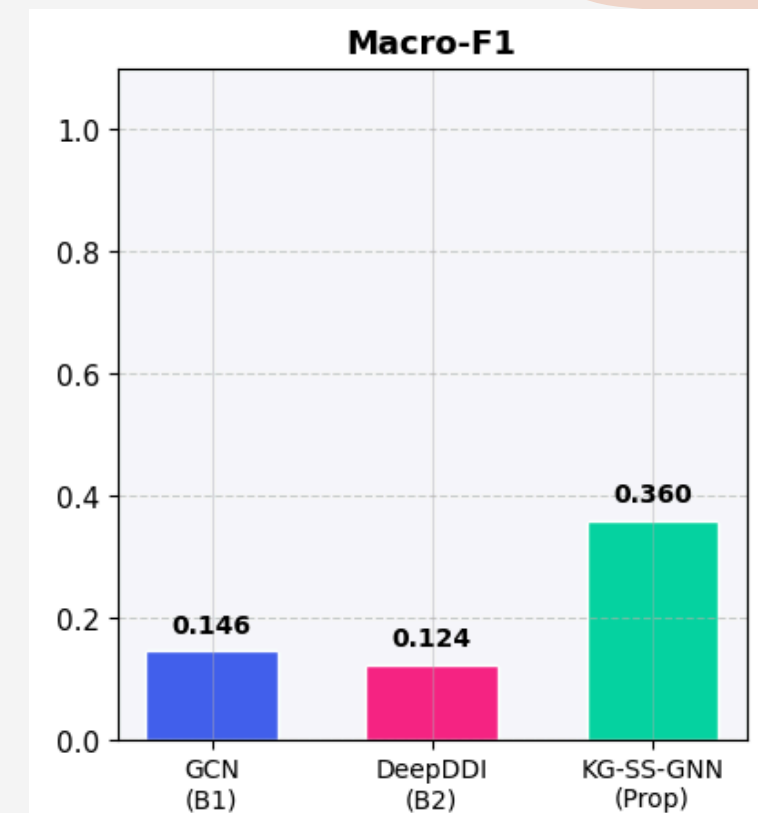
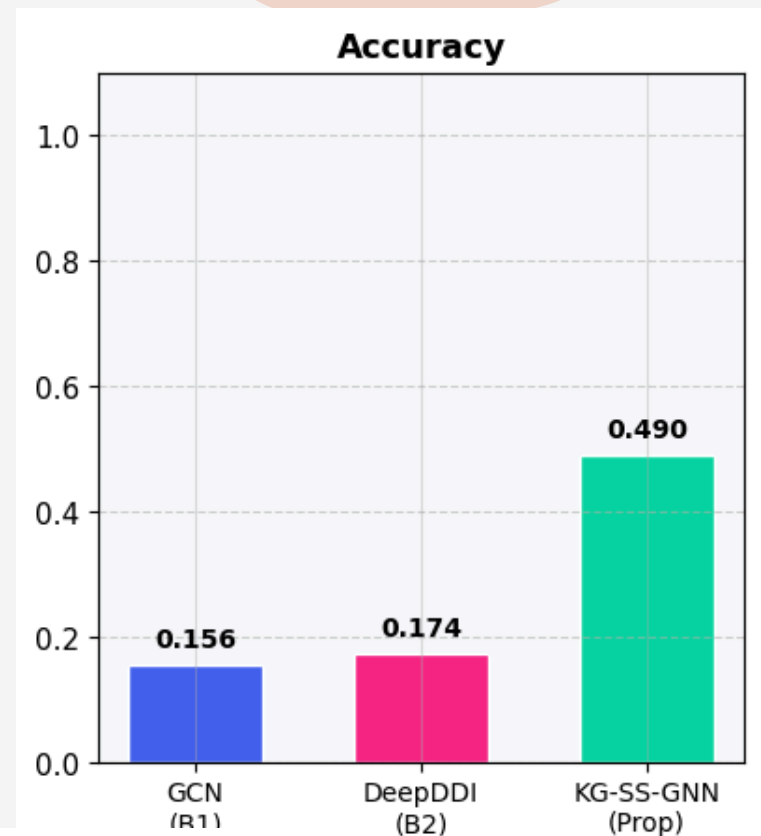


Evaluation Metrics

Metric	Description
Accuracy	Measures the overall percentage of correct predictions
Macro-Recall	Measures how many of the actual interactions the model managed to catch
Macro-F1 Score	Balances catching interactions with avoiding false alarms. The "Macro" version ensures rare interaction types are treated as importantly as common ones.
AUROC	Measures the model's ability to distinguish between an interacting drug pair and a safe drug pair.
AUPRC	Evaluates how well the model finds rare interaction types within a huge dataset of mostly common cases.

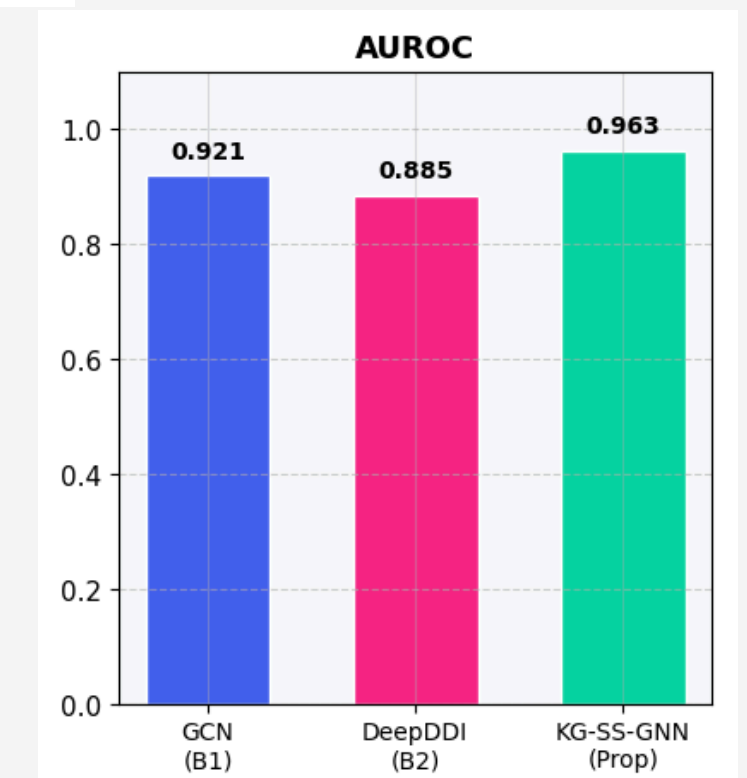
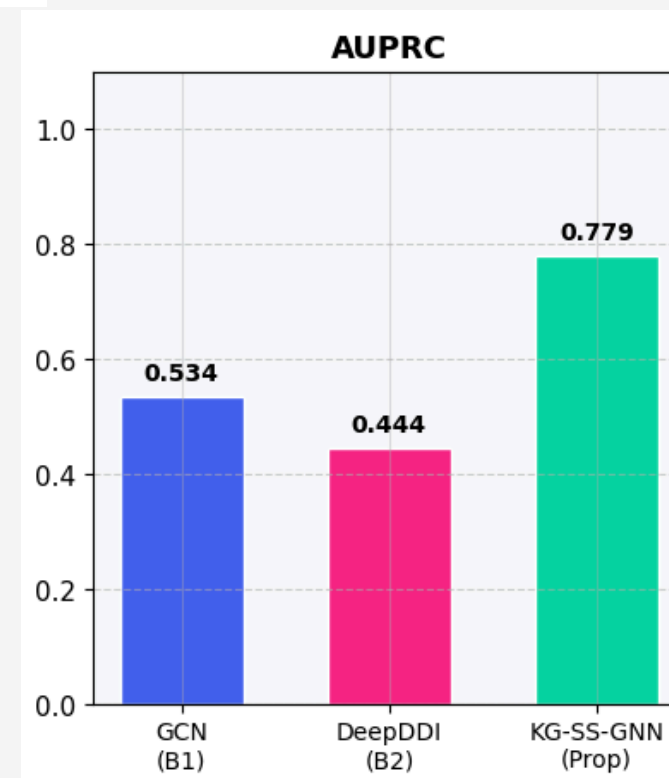
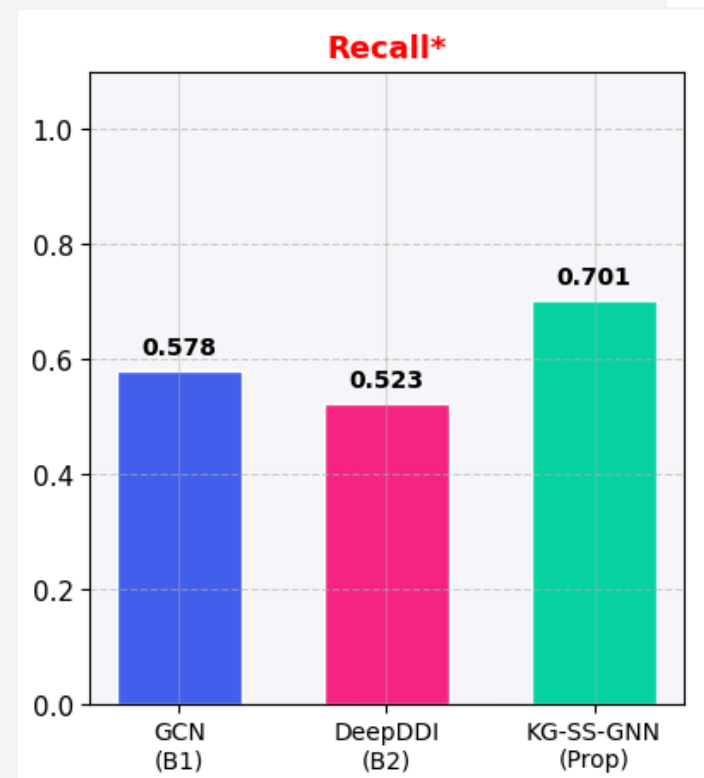
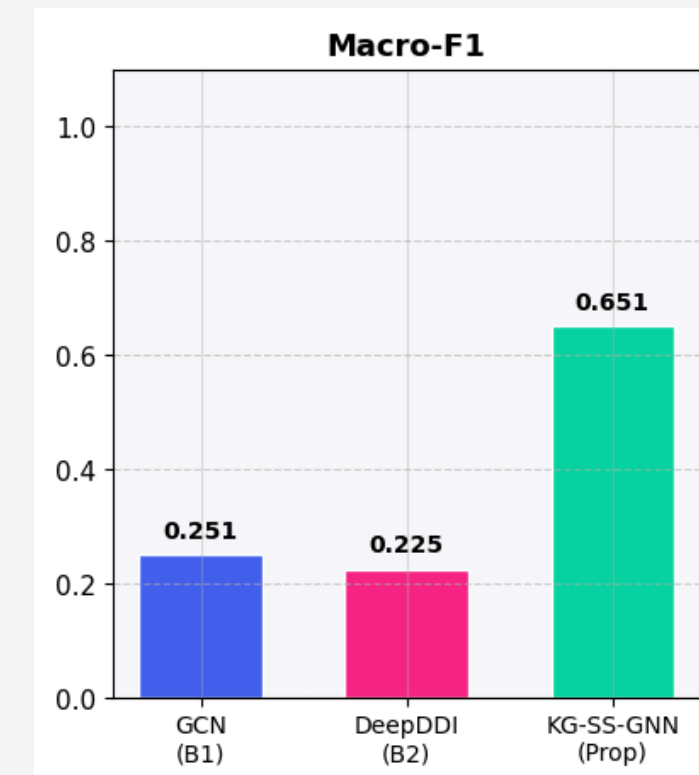
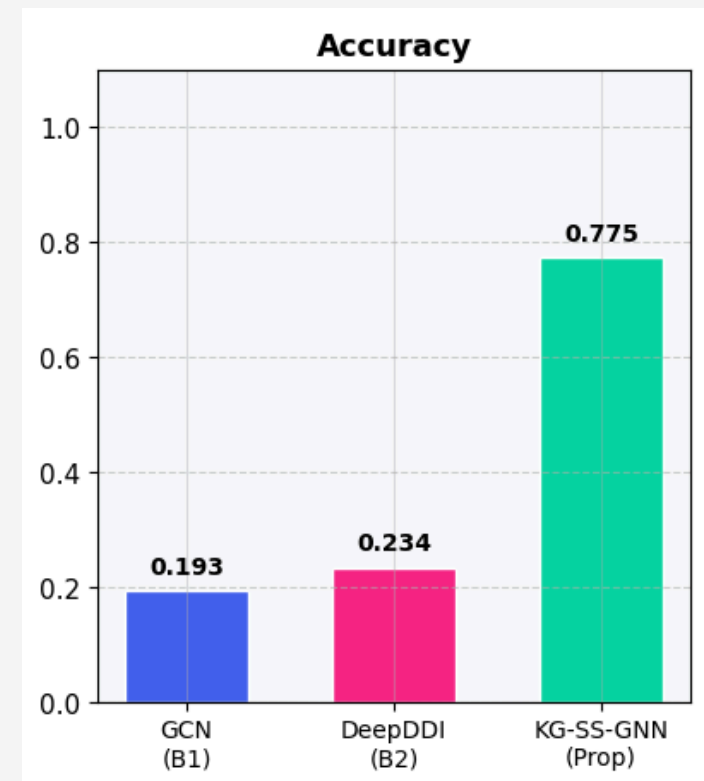
Models Performance on Test Set

Cold Splitting on Both drugs
(unseen drug, unseen drug)

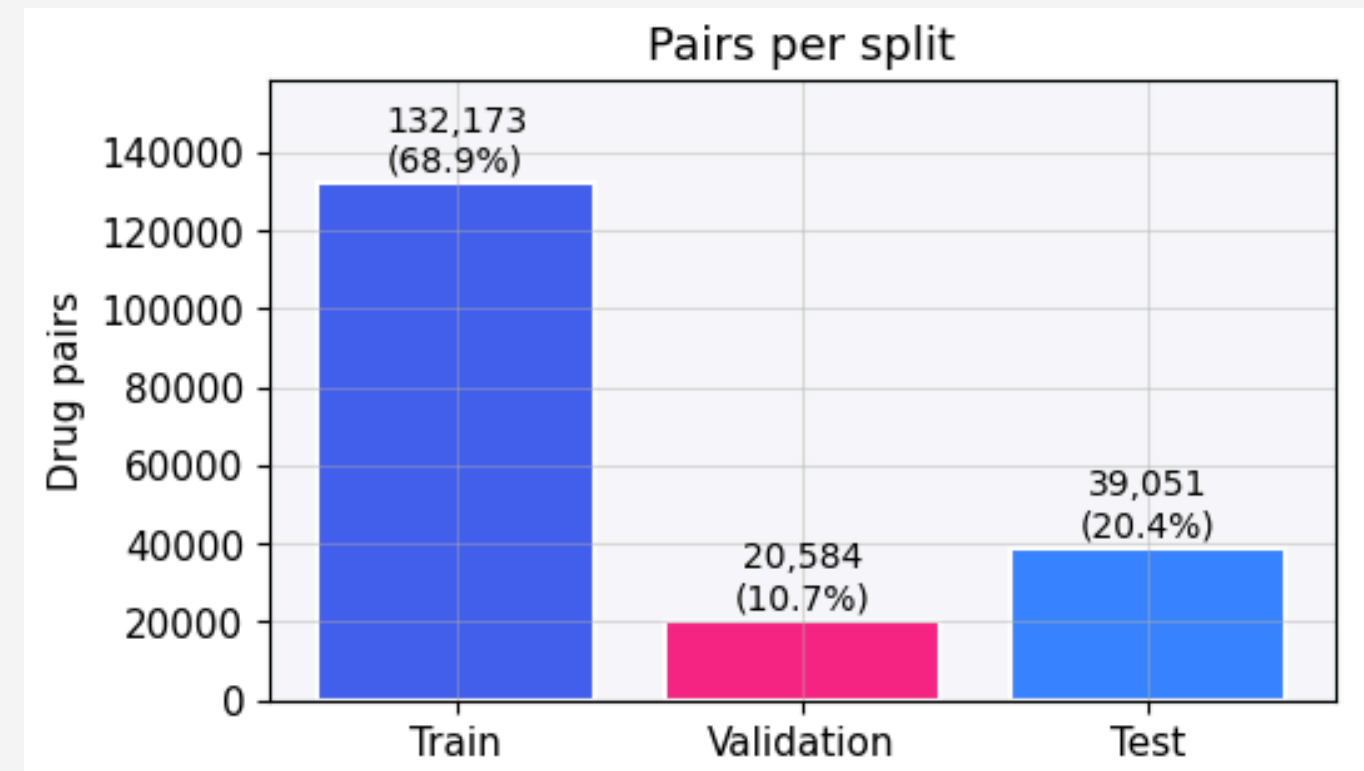
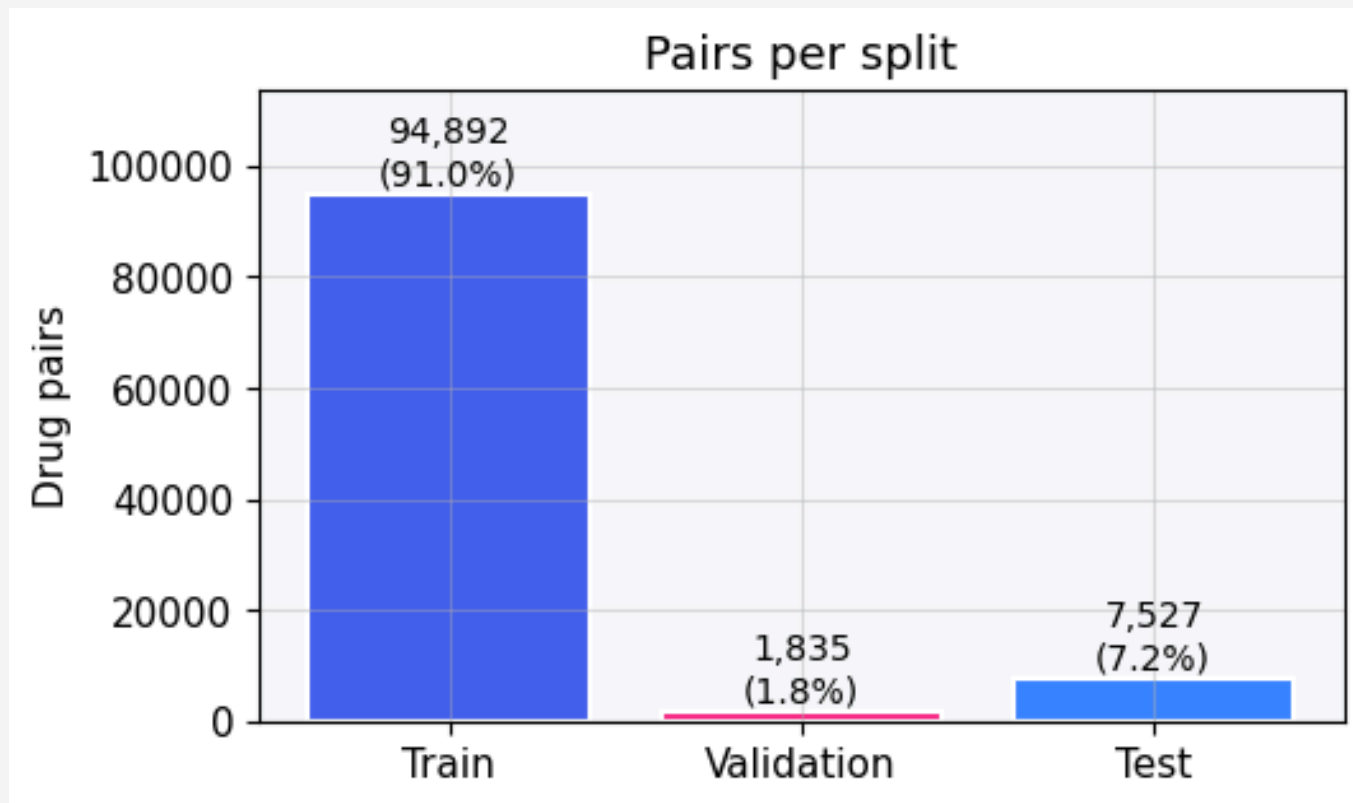


Models Performance on Test Set

Cold Splitting on only ONE drug
(unseen drug, seen drug)



Data Statistics after Splitting



Lessons learned

- Cold Splitting Mechanism
- SMILES Preprocessing
- Handling Imbalanced Data
- Graph Neural Network

Deployment

>>

Deploy ⋮

KG-SS-GNN Drug-Drug Interaction Prediction

Inference demo using the final KG-SS-GNN architecture and local DrugBank SMILES inputs.

Example pair

Custom ▾

Drug A name

Drug A

Drug B name

Drug B

Drug A SMILES

[H][C@@]12C[C@@H](C)[C@](C)(C(=O)CC)[C@@]1(C)C[C@H](O)[C@@]1([H])[C@@]2([H])CCC2=CC(=O)C=C[C@]12C

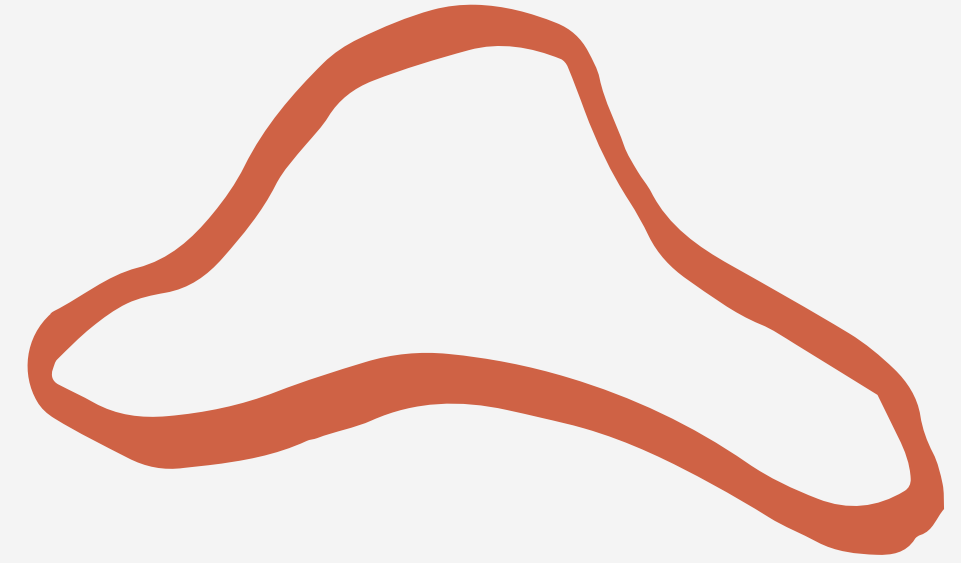
Drug B SMILES

CC(C)NC(=O)NS(=O)(=O)C1=C(NC2=CC=CC(C)=C2)C=CN=C1

Predict interaction

Predictions for Drug A + Drug B

label	probability
Interaction type 82	0.9957
Interaction type 48	0.0017
Interaction type 28	0.0006



THANKS FOR
LISTENING

